

Empirical Regularities in the Intra-Week Trading Patterns of Foreign Institutional Investors in India

Dr. K. N. Badhani*

First draft: March, 2006

Abstract

In developed markets, particularly in US, it has been found that the individual investors use the Monday morning to frame their investment strategies for the coming week. Therefore, their investment activities remain sluggish on this day of the week. This, together with increased selling pressure from the individual investors on Monday, is thought to be partially responsible for the so called 'week-end effect'. Foreign Institutional Investors (FIIs) have occupied an important role in Indian stock market and their behaviour may have significant implication for market dynamics. This paper analyses the trading pattern of FIIs during different days of the week for six calendar years from 2000 to 2005. We find that both the buying and the selling volume of FIIs remain significantly low on Tuesday. The average level of buying volume (adjusted for trend) on Tuesday is about 83 percent of their average daily buying volume for all the days of the week taken together. Similarly, the average level of selling volume on Tuesday is only about 87 percent of the all days' average selling volume. This Tuesday-effect on the volume of FII trading is found robust in the sub-sample-wise analysis. However, no such regularity is observed in the trading patterns of local mutual-funds.

The FIIs, who follow the global investment strategies, possibly use Mondays at their headquarters for strategy formation. Since, it happens after the trading hours in India, their Indian arms may get the instructions only on Tuesday morning and use Tuesdays to frame their local investment strategies. This hypothesis provides a plausible explanation for observed sluggish investment activities of FIIs on Tuesday. This may also explain the significant negative stock returns observed in some countries outside US on Tuesday. We observe that the Tuesday-effect has become weak in the sub-sample period of later three years, which is seems to be a result of gaining more autonomy by agencies managing FII-funds in India.

Key Words: Foreign Institutional Investors, Trading Activities, Day-of-the-Week Effect

JEL classification: G10, G12, G14

* Reader, Department of Commerce, D S B Campus, Kumaun University, Nainital-263002, Uttaranchal. Mobile: 09412908097. E-mail: badhanikn@Yahoo.co.in.

Introduction:

Foreign Institutional Investors (FIIs) have occupied an important place in Indian stock market. As per an estimate of the Economic Times Intelligent Group, the combined ownership of FIIs in BSE500 companies was about \$60 billion as in September 2005, which amounts to nearly 35 percent of free float of these companies (ETIG, 2005). As a result, FIIs do have the capacity to move the market and are thought to be responsible for much of the market volatility we have seen in the recent times. Therefore, an understanding of the behaviour of FIIs is crucial to understand the dynamics of the market. In this paper we analyse the intraweek trading patterns of FIIs in India. We find peculiar regularities in their trading patterns. Their buying and selling volumes tend to be low on Tuesday. We propose that this Tuesday-effect may be a reflection of Monday-effect on institutional investors trading activities documented in US. The FIIs, who follow the global investment strategies, possibly use Mondays at their headquarters for strategy formation. Since, it happens after the trading hours in India, their Indian arms may get the instructions only on Tuesday morning and use Tuesdays to frame their local investment strategies. Consequently, their volume of trade remains low on Tuesday. This hypothesis, although still subject to direct verification, can possibly explain the low Tuesday stock returns documented in many studies outside US.

Survey of the Literature:

The day-of-the-week effect is one of the widely discussed assets pricing anomalies in the literature. More than thirty years ago, Frank Cross (1973) first pointed out differences in returns across weekdays. Since then the day-of-the-week effect has been widely researched across the countries and asset-markets; however, stock market has remained in the focus of the discussion. The references cited by Chang *et al.* (1993), Agrawal and Tandon (1994) and Maberly (1995) provide comprehensive survey of literature on day-of-the-week anomaly. The studies generally conclude that the stock returns on Monday, all in average, are negative and they are significantly lower than the returns on other days of the week. This effect is widely reported from different countries, however, with certain variations. For example, Jaffe and Westerfield (1985) report negative returns on Tuesday in Australia and Japan. Agrawal and Tandon (1994) find that

out of eighteen countries included in their study, in eight countries Tuesday has the lowest average returns. Bildik (2004) and Nath and Dalvi (2004) provide comprehensive lists of research studies who have observed the lowest average daily return for Tuesday. This Tuesday-effect is observed outside US, particularly in emerging markets, and often interpreted as a spill-over of Monday-effect in US stock market (Condoyanni *et al.* 1987).

Several hypotheses have been suggested to explain the day-of-the-week effect, specifically the Monday-effect; for example, settlement procedure hypothesis, information release hypothesis, measurement error hypothesis etc. However, none of the hypotheses could gain unanimous acceptability as they provide only a partial explanation of the anomaly. One of the widely discussed hypotheses proposed to explain the day-of-the-week effect is the *information processing hypothesis* (Miller, 1988; Lakonishok and Maberly, 1990; Abraham and Ikenberry, 1994; Sias and Stark, 1995). This hypothesis is based on the premises that the institutional and individual investors behave differently across the weekdays and the differences in the trading pattern of these two classes of investors is responsible for observed Monday-effect. The crux of this hypothesis is that gathering information during weekdays trading hours is particularly costly for individual investors as most of them are employed with other activities during that period. For them, weekend provide a convenient opportunity to gather and process the information and to reach at an investment decision. They may put some buying orders on weekdays also based on their brokers' recommendations. It has been observed that the broker community produces more buying recommendations than the selling recommendation; hence, for selling decisions the individual investors have to be based on their own information processing. Therefore, there is a clusterisation of selling orders from individuals on Monday (and possibly on Tuesday also).

On the other hand, Osborne (1962) postulates that the institutional investors use Monday morning to frame the trading strategy for the coming week, therefore there is less trading from institutional traders on Monday. This situation produces a downward pressure on prices on that day. Osborne's hypothesis is also supported by empirical evidence. Lakonishok and Maberly (1990) document low volume of block trade on Monday which is associated with trading by institutional investors. They find that Monday is the day with the lowest trading volume. The propensity of individuals to

transact on Monday is highest in relative to other days of the week and that of institutional investors is the lowest. The propensity of individuals to sell on Monday is higher than their propensity to buy. Sias and Stark (1995) explain this weekend behaviour of institutional investors as that of the informed traders in Foster and Vishwanathan (1990) model. Kamara (1997) observes that the Monday seasonal has declined with increasing role of institutional investors in stock market. Chen *et al.* (2004) find that stock portfolios with high institutional holding exhibit a weaker Monday-effect than stock portfolios with low institutional holding. Recently, Venezia and Shapira (2005) have attempted to directly observe the trading behaviour of professional (institutional) and amateur (individual) investors in Israel. Their results also support the previous hypotheses regarding individual and institutional investors' behaviour. They find that weekends influence both amateurs and professional investors; however they affect them in opposite directions. The amateurs increase both their buy and sell activities, and their propensity to sell rises more than their propensity to buy. Professionals on the other hand tend to perform fewer buy as well as sell trades after the weekend, but unlike amateurs, the drop in their activity is almost symmetric for buy trades and for sell trades.

Today the institutional investors have occupied very important role in the stock markets worldwide. They hold major stake in firms' capital and contribute a significant proportion of the trading volume in the market. The increase in the number of institutional investors trading on stock markets worldwide since the end of the 1980s has been associated with a rise in the financial economists' interest in the study of its impact on stock prices. The institutional investors exhibit quite different behaviour in comparison to the individual investors. In particular, there is a popular belief that they destabilize stock prices. Herding and positive feedback trading are the two main arguments put forward for the destabilizing impact on stock prices induced by institutional investors. Therefore, a significant share of research efforts is being directed toward the study of their behaviour (see for example, Lakonishok, Shleifer and Vishny, 1992; Grinblatt, Titman and Wermers 1995; Sias and Starks, 1997; Nofsinger and Sias, 1999; Badrinath and Wahal, 2002; Griffin, Harris and Topaloglu, 2003; Bohl and Brzeszczynski, 2005).

In India, FIIs, for their significant role in the stock market, have attracted considerable attention of the researchers. Studies have mainly focused on the determinants of FII investment flow in India and impact of their behaviour on stock prices and exchange rates (for example, Chakrabarti, 2001; Mukherjee, Bose and Coondoo, 2002; Gordon and Gupta, 2003; Bose and Coondoo, 2004; Badhani, 2005). On the other hand, many studies have examined the intraweek behaviour of the stock prices (for example, Chaudhary, 1991; Poshakwale, 1996; Goswami and Angshman, 2000; Choudhry, 2000; Bhattacharya *et al.*, 2003, Nath and Dalvi, 2004; Badhani and Kavidayal, 2005). However, to our knowledge, no study has so far attempted to link the behaviour of different classes of investors with day-of-the-week behaviour of stock prices in India. Therefore, in this paper we will attempt to link these two issues. The main objective of this study is to examine the trading patterns of FIIs across the days of the week. For comparative purpose we shall also analyse the trading behaviour of another important class of institutional investors – the local mutual funds. In light of our observations we shall also try to explain the possible implications of FII trading behaviour for intraweek dynamics of stock prices.

The Data:

This study is based on a sample period of six years from January 1, 2000 to December 31, 2005. The data on daily FII purchase, sales and net investment have been obtained from the Security and Exchange Board of India. Similar data on investment behaviour of local mutual-funds have also been obtained from the Economic Times, which could be available only for the period of three years from January 1, 2003 to December 31, 2005.

Table: 1 about here

Table: 1 shows the investment pattern of FIIs during different days of the week. The average purchase, sales and net investment for different days of the week have been obtained on the basis of six years daily data. The index of purchase (sales or net investment) on a particular day of the week has been obtained by dividing the mean purchase (sales or net investment) for that day of the week by the corresponding mean purchase (sales or net investment) for all the days of the week taken together. The trading activities of FIIs do not seem to be identically distributed across the days of the week.

The highest average daily purchase reaches on Friday which is 9.1 percent higher in comparison to all-days-average purchase. The lowest average daily purchase reaches on Tuesday which is 11.5 percent below the all-days-average purchase. Similarly the highest level of average daily sales is observed on Friday (8.6 percent below the all-days-average sales) and the lowest level on Tuesday (9.0 percent below the all-days-average sales). The net investment is the difference between daily purchase and daily sales and therefore this shows quite high volatility (it may take both positive and negative values). The highest average net investment is observed on Monday while Tuesday shows the lowest average net investment.

Table: 2 about here

For the analysis of a time series we must ensure that the series is stationary. We use Augmented Dickey-Fuller (ADF) unit-root Test to examine the original and transformed series of FII investment data. Table: 2 presents the results of this analysis. At their original forms (i.e. at level) purchase and sales series are non-stationary; in other words, they are possessing unit roots. However, the net investment series is stationary. Purchase and sales series remain non-stationary at level after their log-transformation (using natural logarithm) also, but the first-differences of log-transformed data, which represent the relative changes (in percentage terms) in purchase and sales respectively, are stationary. Therefore purchase and sales series are ‘integrated of order-1’ (i.e. $I(1)$), while net investment series is ‘integrated of order-0’ (i.e. $I(0)$). We can model the purchase and the sales series using their differenced forms, but this form reflects only the percentage change in the variable on a particular day in relation to the previous day. In this study we are not interested in *percentage change* in FII investment on a particular day rather we want to study the *level* of investment (purchase, sales and net investment) during the different days of the week. For this purpose we compute the *relative levels* of purchase and sales on a particular day in relation to its moving average. We use moving average to de-trend the series. More precisely, first we obtain twenty-days moving average of log-transformed purchase and sales and then subtract it (instead of dividing, as the data are log-transformed) from the corresponding log-transformed purchase or sales of the day. The lag length of the moving average has taken twenty days, which is roughly equivalent to the average number of trading days in a month. The ADF unit root

test (Table: 2) suggests that the relative purchase and the relative sales series are stationary. Therefore, we use these two series for further analysis. The net investment has been used at level without any transformation because it is a stationary series and its log-transformation is not possible because of the presence of negative figures. To examine the robustness of our results, we divide the sample period of six years into two non-overlapping sub-sample periods of three years each. The first sub-sample period covers the duration of three years from January1, 2000 to December 31, 2002 and the second sub-sample period covers the rest of the three years from January1, 2003 to December 31, 2005.

Analysis and Results:

Table: 3 presents the mean relative levels (or *indices*) of FIIs' purchase and sales together with the mean daily net investment on different days of the weeks. We use analysis of variance (ANOVA) to test the hypothesis whether the mean of the variables vary significantly across the days of the week. The independent sample t-test is used to examine whether the mean of the variable for a particular weekday is significantly different from the mean for the rest of the week.

Table:3 about here

For the entire sample period, ANOVA suggests that the mean relative purchase and the mean relative sales vary across the days of the week (F-Ratio is significant at 0.01 level). However, the net investment seems to be identically distributed across the weekdays. The comparison of day's mean relative level of the variable with its mean relative level for the rest of the week using t-test, suggests that the relative level of purchase is significantly low on Tuesday and high on Monday and Friday ($p < 0.01$ in all the cases). The mean relative level of the purchase on Tuesday is about seventeen percent below the mean relative purchase for all the days taken together. Mean relative levels of purchase on Friday and Monday are respectively about ten percent and nine percent higher in comparison to the mean relative purchase for all the weekdays taken together. On the other hand the mean relative level of sales on Tuesday is about twelve percent below than all days mean relative level of sales and this difference is also statistically

significant ($p < 0.01$). The relative level of sales on Friday is also significantly higher ($p < 0.05$) than the mean relative level of the sales for rest of the weekdays.

In the first sub-sample period, ANOVA shows that all the three variables (relative purchase, relative sales and net investment) are not identically distributed across the days of the week ($p < 0.01$ in all the cases). In comparison to mean level for other days of the week, the mean relative level of purchase is significantly lower on Tuesday ($p < 0.01$) and higher on Monday ($p < 0.01$) and Friday ($p < 0.05$). Mean relative level of sales is lower on Tuesday ($p < 0.01$) and higher on Monday ($p < 0.01$). Net investment is also significantly lower on Tuesday and higher on Monday ($p < 0.05$ in both the cases). However in the second sub-sample period net investment is not found significantly different across the days of the week. The mean relative purchase and the mean relative sales are showing significant variations across the days of the week ($p < 0.01$). Both are significantly lower on Tuesday and higher on Friday in comparison to their respective mean levels for rest of the weekdays. However, we do not find any significant variation in the level of relative purchase or sales on Monday in second sub-sample period.

To sum-up, we can say that the trading activities (both purchase and sales) of FIIs are found sluggish on Tuesday. Although we also find evidence suggesting that the levels of FII-purchase and sales remain up on Monday and Friday, but this trend is not found robust across the sub-samples. The deviations in the mean levels of purchase and sales on Tuesday are much sharper during the first sub-sample period (26 and 18 percent respectively) in comparison to the second sub-sample period (9 and 7 percent respectively). It is also notable that whatever deviation we observe on a particular weekday, that is in the same direction in purchasing and selling activities of FIIs. Consequently, these deviations do not cause much disturbance in the level of net FII investment.

Financial time series are known for showing persistence of trend and volatility. Sometimes it has been found that so-called calendar anomalies become insignificant when adjustment is made for such persistence (Lian and Chen, 2004). Therefore, to take care of this, in a more rigorous econometric analysis we use dummy variable regression models of relative purchase and relative sales with conditional mean and variance specification. Dummy variables in the models represent five days of the week. Five

autoregressive (AR) terms have been used to adjust for conditional mean and the variance has been specified as GARCH (1, 1) process. The choice of five AR terms is rather arbitrary, but it has been motivated with the objective to make the estimation of conditional mean smooth and unbiased for the weekdays through taking into consideration all the five trading days of the week. More specifically, the following regression models have been used;

$$\begin{aligned}
P_t &= \sum_{i=1}^5 \alpha_i P_{t-i} + \sum_{j=1}^5 \beta_j D_j + \varepsilon_t \\
\Rightarrow \varepsilon_t / \varepsilon_{t-1} &\approx N(0, \sigma^2) \\
\sigma^2 &= \omega + \gamma_1 \varepsilon_{t-1} + \gamma_2 \sigma_{t-1}^2
\end{aligned}
\tag{1}$$

and similarly;

$$\begin{aligned}
S_t &= \sum_{i=1}^5 \alpha_i S_{t-i} + \sum_{j=1}^5 \beta_j D_j + \varepsilon_t \\
\Rightarrow \varepsilon_t / \varepsilon_{t-1} &\approx N(0, \sigma^2) \\
\sigma^2 &= \omega + \gamma_1 \varepsilon_{t-1} + \gamma_2 \sigma_{t-1}^2
\end{aligned}
\tag{2}$$

Where, P_t and S_t represent the relative purchase level and relative sales level respectively. D_j ($j=1, \dots, 5$) are the days of the week dummies. The intercept has not been used to avoid dummy variable trap.

Table: 4 about here

Table: 4 presents the results of this analysis (only coefficients of dummy variables have been reported). These results reinforce our earlier inferences based on ANOVA and independent sample t-tests. We observe a consistent and significant shift in conditional mean of the relative purchase and the relative sales on Tuesday for entire sample as well as for the sub-sample periods. Any other day-of-the-week effect is not observed with such a high level of consistency. After making adjustment for conditional mean and variance we observe that on Tuesday the relative purchase and the relative sales are depressed by about 21 percent and 14 percent respectively during the entire sample period. The results also support our earlier conclusion that the negative Tuesday-effect on FIIs trading activities is stronger during the first sub-sample period in comparison to the

second sub-sample period. The downward shift in the conditional mean of the relative purchase is about 30 percent during the first sub-sample period on Tuesday while during the second sub-sample period it moves down about 12 percent only. Similarly the conditional mean of relative sales moves down by about 19 percent and 15 percent respectively during the first and the second sub-sample periods. During the first sample period as well as for the entire sample period we observe a significant upward shift in the conditional mean of relative purchase on Monday and Friday but in the second sub-sample the Monday-effect reverses its sign and turns negative (although statistically insignificant). We also observe a significant upward shift in conditional mean of relative sales on Monday during first sub-sample period but this also turns negative and insignificant during the second sub-sample period.

Table: 5 about here

We have also the data on trading activities of local mutual funds for the period of three years corresponding to our second sub-sample period. We analyse these data using similar procedure as we used for analysis of FII investment activities. The results (Table: 5, Panel: A&B) do not indicate presence of any robust day-of-the-week effect in their investment behaviour. ANOVA indicates presence of day-of-the-week effect in relative sales ($p < 0.05$). Independent sample t-test shows the presence of a positive Tuesday-effect on both the relative purchase and the relative sales ($p < 0.01$); but after making adjustment for conditional mean and variance, only positive Tuesday-effect on relative purchase remains significant at 0.05 level. Unlike to FII trading activities no Tuesday-effect is observed in trading activities of local mutual funds; therefore, it seems that the sluggish trading activity on Tuesday is a unique feature of FIIs.

Conclusions:

In this paper we have attempted to detect day-of-the-week effect on the trading activities of FIIs in India. Although, net FII investment seems to be distributed identically across the days of the week, but relative levels of purchase and sales have been found significantly low on Tuesday. A parallel analysis of the trading activities of local mutual funds shows that their trading activities are almost identically distributed across the weekdays and there is no abnormal deviation in them on Tuesday. In another study, the analysis of trading volume at the National Stock Exchange of India (NSE) shows that the

overall trading volume remains depressed on Monday rather than on Tuesday (Badhani and Kavidayal, 2005). Therefore, the sluggish trading activity on Tuesday seems to be a unique feature of FII investment behaviour.

To find possible explanations of this Tuesday-effect on FII trading activities we may look into the market microstructure and practices. Tuesday used to have special significance under the fixed day weekly settlement system, as it was the settlement day at NSE. But there is little possibility that this fact is associated with sluggish trading activities of FIIs on this day of the week. The fixed day weekly settlement system has been replaced by the rolling settlement system since January 1, 2002; but, the Tuesday-effect on FII trading activities is observed consistently even after January, 2002. Moreover, we expect increased, rather than decreased, trading activities on the settlement day under the fixed day weekly settlement system as on this day the investors who have taken speculative positions during the week are likely to square-off their positions, particularly when carry-forward is not allowed. But in contrary to this expectation, we observe sluggish trading activities on Tuesday uniquely associated with FIIs.

It seems that the Tuesday-effect is a reflection of Monday-effect on institutional investors trading activities found in US. As it has been discussed earlier, studies shows that the volume of trading of institutional investors remains low on Monday because they use Monday morning to frame their investment strategies for the coming week. In India, most of the FIIs are foreign mutual-funds (recently some pension funds have also got registered themselves as FIIs). They manage the portfolio of their own or of the sub-account of other foreign investors (in majority of the cases, institutional investors). Domestic asset management companies may also work as FIIs for managing the portfolio of the sub-accounts of foreign investors. These FIIs are likely to get regular instructions from their head quarters or from the sub-account holders regarding the investment strategies to be followed. It is possible that the foreign institutional investors, who follow the global investment strategies, use Mondays at their headquarters for strategy formation. Since, it happens after the trading hours in India, the asset management agencies (foreign or domestic) in India may get the instructions only on Tuesday morning. Therefore, they are likely to use Tuesday to frame their local investment strategies, and their trading activities remain sluggish on this day. Consequently, the

Monday-effect documented in US regarding the institutional investors' behaviour is postponed till Tuesday in case of FIIs in India.

This hypothesis, although still subject to direct verification, provides a plausible explanation for observed sluggish investment activities of FIIs on Tuesday. It may also throw some light why in some countries outside US the day-of-the-week effect on stock market returns is observed on Tuesday instead of Monday. In India also we observe a consistent (but statistically not significant for recent time period) downward shift in S&P CNX Nifty returns on Tuesday (Badhani and Kavidayal, 2005). Many studies document the spill-over effect of stock market trends and volatility from US to the emerging markets (see Biswal and Mohanty, 2003 for references) and the Tuesday-effect is also explained as a spill-over effect of Monday-effect in US. Possibly, the behaviour of FII investments as documented in this study is one of the channels of linkages among international stock markets and may be helpful, to some extent, in explaining the sources of spill-over effects.

The decreasing intensity of Tuesday-effect in second sub-sample period in comparison to first sub-sample period may be an indication of increasing independence gained by the agencies managing FII funds in India in strategy formation. It seems that FIIs are giving more autonomy to their Indian arms with their increasing experience in Indian market. Consequently, the asset management agencies in India have not to wait till Tuesday for instructions; rather they frame their weekly strategies on Monday only like institutional investors abroad. The behaviour of FII trading volume on Monday across the sub-sample periods also supports this hypothesis. Monday exhibits significantly high level of FII trading (both purchase and sales) during the first sub-sample period but in the second sub-sample period the level of FII trading is below the weekly average.

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Table: 1
Intra-Week Trading Pattern of FIIs: Average Daily Trading for Different Days of the Week

Day of the Week	Purchase		Sales		Net Investment	
	Average (Rs. Million)	Index	Average (Rs. Million)	Index	Average (Rs. Million)	Index
All Days	4904.9	100.00	3976.1	100.00	928.8	100.00
Monday	5128.3	104.55	3936.1	98.99	1192.1	128.34
Tuesday	4391.0	89.52	3618.4	91.00	772.6	83.18
Wednesday	4779.2	97.44	3911.0	98.36	868.1	93.46
Thursday	4880.3	99.50	4099.3	103.10	781.0	84.09
Friday	5353.2	109.14	4317.5	108.59	1035.7	111.51

Table: 2
ADF Unit Root Test

Test Statistics (τ)

Variable	Null Hypothesis	
	Unit Root	Unit Root with Drift
1. Purchase	0.45	2.30
2. Sales	0.71	2.44
3. Net Investment	6.44***	7.19***
4. Log Purchase	0.97	2.55
5. Log Sales	2.24	3.96*
6.Change in Purchase	10.39***	10.48***
7. Change in sales	12.29***	12.35***
8.Relative Purchase	8.86***	9.01***
9.Relative Sales	10.46***	10.55***

*** $p < 0.01$

** $p < 0.05$

* $p < 0.10$

Table: 3
Day-of-the-Week Effect on FII Investment: Preliminary Analysis

Day of the Week	Relative Purchase (Index)		Relative Sales (Index)		Net Investment (Rs. Milloan)	
	$\bar{X}$	t	$\bar{X}$	t	$\bar{X}$	t
Entire Sample (2000-2005)						
All Days	0.0037	-	0.0030	-	928.8	-
Monday	0.0905	3.71***	0.0382	1.47	1192.1	1.94*
Tuesday	-0.1732	7.72***	-0.1255	5.42***	772.6	1.16
Wednesday	0.0067	0.12	0.0086	0.23	868.1	0.45
Thursday	-0.0050	0.38	0.0332	1.27	781.0	1.10
Friday	0.1036	4.12***	0.0620	2.42**	1035.7	0.78
F-Ratio	18.55***		7.83***		1.24	
First Sub-sample (2000-2002)						
All Days	-0.0075	-	-0.0090	-	307.6	-
Monday	0.1851	5.62***	0.1333	4.01***	546.6	2.48**
Tuesday	-0.2628	7.50***	-0.1827	4.88***	98.7	2.16**
Wednesday	0.0018	0.26	-0.0032	0.16	277.5	0.31
Thursday	-0.0415	0.98	0.0073	0.46	146.1	1.69*
Friday	0.0805	2.46**	-0.0018	0.19	476.7	1.71*
F-Ratio	19.77***		8.11***		3.37***	
Second Sub-sample (2003-2005)						
All Days	0.0148	-	0.0147	-	1549.2	-
Monday	-0.0040	0.60	-0.0569	1.73*	1842.0	1.19
Tuesday	-0.0872	3.31***	-0.0705	2.81***	1428.2	0.46
Wednesday	0.0115	0.11	0.0204	0.18	1466.6	0.38
Thursday	0.0317	0.64	0.0592	1.40	1428.6	0.46
Friday	0.1261	4.06***	0.1241	3.86***	1587.1	0.15
F-Ratio	4.90***		5.29***		0.41	

*** p< 0.01

** p< 0.05

* p< 0

Table: 4
Day-of-the-Week Effect on FII investment: Dummy variable Regression with
Conditional Mean and Variance Specification (AR-5, GARCH-1, 1)

(Beta-coefficients of the day-of-the-week dummies)

Day-of-the-Week	Relative Purchase		Relative sales	
	β_j	t	β_j	t
Entire Sample (2000-2005)				
Monday	0.0534	2.08**	0.0030	0.14
Tuesday	-0.2063	8.68***	-0.1401	5.91***
Wednesday	0.0311	1.05	0.0220	0.76
Thursday	0.0112	0.40	0.0530	1.86*
Friday	0.1019	3.59***	0.0446	1.49
First Sub-sample (2000-2002)				
Monday	0.0663	3.82***	0.1408	5.57***
Tuesday	-0.3028	8.71***	-0.1872	6.07***
Wednesday	0.0212	0.50	0.0166	0.47
Thursday	-0.0105	0.27	0.0472	1.43
Friday	0.0846	2.08**	-0.0446	1.04
Second Sub-sample (2003-2005)				
Monday	-0.0475	1.48	-0.0030	0.14
Tuesday	-0.1160	3.33***	-0.1501	6.33***
Wednesday	0.0325	0.77	0.0220	0.76
Thursday	0.0433	1.04	0.0430	1.51
Friday	0.1326	3.53***	0.0446	1.49

*** p< 0.01

** p< 0.05

* p< 0.1

Table: 5
Day-of-the-Week Effect on Investment Behaviour of Local Mutual-Funds

Panel-A: Preliminary Analysis

Day of the Week	Relative Purchase (Index)		Relative Sales (Index)		Net Investment (Rs. Milloan)	
	$\bar{X}$	t	$\bar{X}$	t	$\bar{X}$	t
Entire Sample (2003-2005)						
All Days	0.0114	-	0.0109	-	16.94	-
Monday	-0.0320	1.58	-0.0133	0.96	10.02	1.01
Tuesday	-0.0392	1.44	-0.0118	0.62	10.36	0.97
Wednesday	0.0270	0.60	0.0415	1.20	16.50	0.06
Thursday	0.0861	2.62***	0.0878	2.66***	17.02	0.01
Friday	0.0165	0.15	-0.0498	1.85*	31.16	1.82*
F-Ratio	2.20*		2.53**		1.10	

Panel-B: Dummy Variable Regression Model

(Beta-coefficients of the day-of-the-week dummies)

Day of the Week	Relative Purchase (Index)		Relative Sales (Index)	
	β_j	t	β_j	t
Entire Sample (2003-2005)				
Monday	-0.0380	0.96	0.0377	0.92
Tuesday	-0.0294	0.70	-0.0095	0.19
Wednesday	0.0338	0.76	0.0387	0.91
Thursday	0.0779	2.13**	0.0497	1.43
Friday	-0.0075	0.23	-0.0730	1.83*

*** p< 0.01

** p< 0.05

* p< 0.10

